Spectral Co-Design of Agrivoltaic Modules for Multifunctional Net Energy Benefit: A Quantum-Guided Digital Twin

Steve Cabrel Teguia Kouam[a,*], Theodore Goumai Vedekoi[b], Jean-Pierre Tchapet Njafa[b], Jean-Pierre Nguenang[a], Serge Guy Nana Engo[b]

[a] *Department of Physics, Faculty of Science, University of Douala, Cameroon*
[b] *Department of Physics, Faculty of Science, University of Yaoundé I, Cameroon*

Abstract

Static semi-transparent agrivoltaic modules settle the food–energy conflict by paying both sides: fractional electrical output for fractional crop light. We ask whether engineering the transmitted spectrum at the material level, instead of dimming it uniformly, turns that fixed ratio into a design variable. The stack couples a dual-band (750 nm and 820 nm) organic photovoltaic filter to a plasmonic nanoparticle-on-mirror (NPoM) cavity confined to 1 % of the module area, which converts the cavity's 91.8 % local excitation-transport penalty into a multiplexed SERS and quantum-dot diagnostic channel for pesticides, heavy metals and crop stress — 0.9 % of canopy transfer yield (0.980 to 0.971, a device-level metric) for the sensing function. The same passband prepares polaron-dressed states in the Fenna–Matthews–Olson complex and extends exciton coherence by 50 %; no revenue line in the accounts passes through that quantum gain. At field scale the shield cuts crop evapotranspiration by 28 % (FAO-56, 460 $\mathrm{L\,m^{-2}\,yr^{-1}}$ water credit) and returns a Net Ecological Benefit of 72.6 $\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$ for a modeled 500 $\mathrm{m^2}$ cooperative module in Cameroon, with the payback of an unsubsidized 6.17 yr shortened to 4.32 yr by 30 % blended finance and 2.68 yr by local-material construction, over telemetry secured by BB84 key distribution. What binds the design is not economics, which existing instruments already close, but process engineering: sub-2 nm cavity gaps, sentinel fouling, and organic-photovoltaic encapsulation lifetime.

Keywords: agrivoltaic spectral management, net ecological benefit, FAO-56 microclimate, plasmonic nanoparticle-on-mirror cavity, SERS diagnostics,

*Corresponding author. Email: steve.teguia@facsciences-uy1.cm

techno-economic analysis, quantum-guided digital twin

1. Introduction

Agrivoltaics co-locates photovoltaic (PV) systems and agricultural cultivation to mitigate the land-use competition between food production and renewable energy generation [1]. Its land-use optimization and mobile-PV foundations are well documented [2, 3], and recent syntheses systematize overhead, vertical, and tracking typologies with land-equivalent-ratio and levelized-cost metrics [4]. Across these studies, three gaps recur: (i) the optical architecture is static, so spectral light-sharing is not co-optimized with crop photophysiology [5, 6]; (ii) techno-economic and lifecycle accounts remain decoupled from the underlying transport physics [7, 8]; and (iii) the PV layer is treated purely as an energy harvester, with crop-stress and contaminant sensing bolted on as separate hardware [9, 10, 11]. Dynamic, physics-guided spectral management and multifunctional co-design have not yet been transposed into a single operating agrivoltaic stack.

The photosynthetic apparatus of green plants and bacteria has evolved under a spectrally structured photon flux, where specific wavelengths drive charge separation while others dissipate as heat [12]. This spectral specificity lets the PV layer transmit only the photosynthetically active wavelengths while absorbing the remainder, converting the food-energy conflict into a problem of spectral co-design.

The Fenna–Matthews–Olson (FMO) complex of green sulfur bacteria serves as the model system for spectral light-sharing [13]. Its seven bacteriochlorophyll *a* sites have been extensively characterized through two-dimensional electronic spectroscopy, revealing oscillatory coherence signatures that persist for hundreds of femtoseconds at cryogenic [14] and room temperature [15]. Although the functional relevance of these quantum beats remains debated, consensus has emerged that vibronic coupling — the hybridization of electronic transitions with underdamped protein vibrations — governs the flow of excitation energy [16, 17]. Non-Markovian frameworks such as the hierarchical equations of motion [18] and the process tensor hierarchy of pure states (PT-HOPS) [19, 20] capture these effects with demonstrated numerical convergence. Device-adjacent uses of exact open-quantum-system methods remain rare: the closest precedent is a general HEOM library applied to light-harvesting and control problems without a photovoltaic prod-

uct [21]. Photonic-quantum solar design and vibronic-screening workflows likewise stop short of an operating device [22, 23].

Previous work demonstrated that spectral filtering of the excitation pulse — aligning its spectrum with underdamped vibronic resonances at 750 nm and 820 nm — extends coherence lifetimes by 20 % to 50 % in the 7-site FMO complex [24]. This selective vibronic excitation prepares polaron-dressed states that decouple from the dissipative protein-solvent bath. That companion paper, published in the *Journal of Physical Chemistry Letters*, reports the analysis of the filter, the excitation-transport rate, and the single-pass conversion efficiency [24]; the present work adds the 8-site trapping model, the monolithic OPV–NPoM hardware concept, the SERS field calibration, and the Net Ecological Benefit and lifecycle assessment developed below. Single-photon experiments have confirmed that quantum coherence persists in individual light-harvesting complexes under ambient conditions [25], and quantum phase synchronization via exciton-vibrational energy dissipation has been identified as a mechanism sustaining long-lived coherence [26]. Here we extend this principle along the chain from molecule to module. An 8-site FMO model with a non-Hermitian reaction-center trap on sites 3 and 4 turns the spectral-selection rule into a computable quantity, the time-integrated forward transfer yield Φ_{FT}. Embedding the complex in a nanoparticle-on-mirror (NPoM) plasmonic cavity [27] couples the biological light-harvesting system to the organic photovoltaic (OPV) layer through strong coherent coupling, forming a 9×9 dressed Hamiltonian in which the same cavity that gates energy flow also serves as the SERS nanoreactor. Floquet–Stark detuning and flux-dependent optical limiting then supply a control law that mimics the non-photochemical quenching used by plants under high light stress.

We map this microscopic optimization onto a field-scale lifecycle assessment. We parameterize crop evapotranspiration using the FAO-56 Penman–Monteith model [28] under the spectrally selective OPV shield, compute the Net Ecological Benefit (NEB) across three scenarios, and evaluate socioeconomic access through cooperative ownership models and BB84 quantum key distribution [29] for secure IoT telemetry. This multi-scale perspective [30] links angstrom-scale vibronic couplings to square-meter canopy yields, providing a unified modeling framework for quantum-guided agrivoltaics (Fig. 1).

The central question of this work is falsifiable: *Can spectral co-design guided by non-Markovian quantum dynamics deliver a positive multifunctional net energy benefit (power + water + carbon + diagnostics) in a field-*

constrained agrivoltaic stack, and which barriers remain physical rather than economic? Three subordinate questions structure the Results. (Q1) Device asks what the NPoM co-design costs and where the cost lands, quantified by the mode-volume scan (Table 1) and located by the canopy-integration analysis. (Q2) Physics asks which microscopic states carry the trade-off: exact PT-HOPS dynamics localize the loss channels, and Floquet–Stark detuning with flux-dependent optical limiting specifies a control law under which they could be managed dynamically (its yield-level validation, Φ_{FT} with the switch engaged vs. disengaged, is an Outlook step). (Q3) System asks whether the stack pays for itself, once coupled FAO-56 microclimate, cooperative techno-economics, and BB84-secured telemetry internalize every function into a single Net Ecological Benefit account. Three falsifiable claims anchor the answer. The first, (F1) Microscopic, is that dual-band spectral filtering at 750 nm and 820 nm prepares polaron-dressed states that extend exciton coherence by 50 % and raise the forward transfer yield by 25 % at 295 K — both established in the companion 7-site study [24] and carried over here under the weak site-8 coupling assumption. The second, (F2) Device, is that confining the NPoM cavity to a 1 % sentinel fraction preserves 97.1 % of the global canopy trapping yield ($\Phi_{\mathrm{FT}}^{\mathrm{global}} = 0.971$) while retaining single-complex SERS diagnostics, despite the > 90 % local transport penalty quantified across the mode-volume scan. The third, (F3) System, is that the FAO-56 smart shield cuts evapotranspiration by 28 % (± 10 % microclimate uncertainty) and, combined with OPV generation and cooperative economics, yields a Net Ecological Benefit of $72.6\,\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$ and a 4.32 yr payback at 30 % subsidy for the modeled $500\,\mathrm{m^2}$ cooperative case. We answer affirmatively but conditionally: the bounds (single-exciton treatment, site-independent baths, modeled fouling and matrix penalties, ± 10 % microclimate band) are stated explicitly in the Limitations section of the Discussion. The headline outcome, pictured in Figure 2, compresses this answer to one sentence: a 91.8 % local transport penalty confined to a 1 % sentinel area costs 0.9 points of canopy-scale trapping yield (0.980 to 0.971) while adding single-complex SERS diagnostics to the same footprint. To keep a stack of this breadth auditable, each layer — quantum dynamics, microclimate, economics, and security — is validated separately in the Supplementary Material (Section S2–Section S12), and every headline number below is traceable to a single figure or table.

The value chain of the quantum layer is worth stating explicitly, because the energy-systems reader is entitled to ask what the quantum hardware buys. It contributes three concrete functions: (i) the dual-band passband

co-design that the FAO-56 and lifecycle accounts price into the module's transmittance budget; (ii) multiplexed SERS and quantum-dot diagnostics delivered on the 1 % sentinel footprint at no additional canopy area; and (iii) preservation of 0.971 of the device-level excitation-transfer yield despite the local plasmonic penalty. On the cost side, the NPoM sentinel and the BB84 link appear as capital and operating line items in the techno-economic account. The transport gain itself (+25 % forward transfer yield) is deliberately not monetized: no coherence-derived revenue enters the NEB or payback chains.

2. Results

2.1. Quantum-enabled spectral co-design and NPoM sentinel

The spectral co-design rests on an 8-site FMO model with a non-Hermitian reaction-center trap on sites 3 and 4 (full Hamiltonian, site energies, and 28-pair coupling matrix in Section S1). Dual-band transmission at 750 nm and 820 nm prepares polaron-dressed states that extend the l_1-norm coherence lifetime from 280 ± 25 fs (broadband) to 420 ± 35 fs (filtered) — a 50 % increase — and raise the forward transfer yield Φ_{FT} from 0.71 ± 0.02 to 0.89 ± 0.03 (Figure 3; PT-HOPS/SBD propagation details and convergence in Section S2).

The OPV–FMO interface is mediated by a nanoparticle-on-mirror (NPoM) cavity that forms a 9×9 dressed Hamiltonian. Mode-volume scans (Table 1 and Section S11) show that strong plasmon–exciton coupling suppresses local transport to $\Phi_{\mathrm{FT}}^{\mathrm{NPoM}} \approx 0.080$ at $V_{\mathrm{mode}} = 1.2\,\mathrm{nm}^3$ while enabling relative Purcell-scaled SERS enhancement factors of 100 at $V_{\mathrm{mode}} = 0.8\,\mathrm{nm}^3$ down to 44 at $V_{\mathrm{mode}} = 1.2\,\mathrm{nm}^3$ (a factor-48 enhancement range across the full seven-point scan, from 33 to 1.6×10^3; tabulated endpoints here, full scan in Section S11), consistent with absolute enhancements $\geq 10^6$ reported for sub-nm NPoM gaps. The integrated transmitted power in the two $100\,\mathrm{cm}^{-1}$ passbands is $\sim 10.2\,\mathrm{W\,m}^{-2}$ (~ 1.0 % of AM1.5G), priced into the module transmittance budget. The 91.8 % local suppression relative to the passive baseline ($\Phi_{\mathrm{FT}}^{\mathrm{passive}} = 0.980$) is the price of the diagnostic channel; it is confined to a sentinel area fraction $\alpha = 0.01$.

That budget is what the design can claim on the crop side: the 750 nm and 820 nm windows sit outside the photosynthetically active range, and greenhouse measurements under translucent modules show crop performance

tracking PAR transmittance ahead of every other module optical parameter [31]. The quantum channel therefore buys its diagnostics out of the near-infrared that even plant-transparent designs divert or discard, not out of the photons the canopy harvests. The corresponding land-use ledger, computed from the modeled crop and power shares with the conventional OPV absorption of the remaining flux left as a transmittance choice, is reported in the Supplementary Material.

Table 1: **NPoM mode-volume sensitivity of forward transfer yield (selected points).** Passive baseline $\Phi_{\mathrm{FT}} = 0.980$ (assumed capture-efficiency cap, a model input rather than a trajectory integral). Full seven-point scan, SERS enhancement factors, and n-trajectory provenance in Section S11 ($n = 100$ at $V_{\mathrm{mode}} = 0.8\,\mathrm{nm}^3$, $n = 20$ at $V_{\mathrm{mode}} = 1.2\,\mathrm{nm}^3$, $n = 2$ otherwise). Production parameters: $L = 8$, $K = 2$, $\Delta t = 0.2\,\mathrm{fs}$, $T = 295\,\mathrm{K}$.

V_{mode} (nm^3)	Φ_{FT}	Δ vs. baseline (%)
0.2	0.0505	−94.8
0.8	0.0768	−92.2
1.2	0.0799	−91.8
1.4	0.0791	−91.9

2.2. Global canopy yield integration

Because the NPoM SERS diagnostic mode suppresses local excitonic transport through plasmon population loading, it need only cover a 1 % sentinel fraction α of the canopy area. The remaining 99 % operates in passive OPV mode, preserving the near-baseline forward transfer yield $\Phi_{\mathrm{FT}}^{\mathrm{passive}} = 0.980$. The area-weighted *global* canopy trapping yield is therefore:

$$\Phi_{\mathrm{FT}}^{\mathrm{global}} = \alpha\,\Phi_{\mathrm{FT}}^{\mathrm{NPoM}} + (1-\alpha)\,\Phi_{\mathrm{FT}}^{\mathrm{passive}}, \tag{1}$$

yielding $\Phi_{\mathrm{FT}}^{\mathrm{global}} = 0.971$ at $\alpha = 0.01$ and $\Phi_{\mathrm{FT}}^{\mathrm{NPoM}} = 0.080$ (or 0.0799 precisely); device-level excitation transfer is thus maintained near baseline even though transport is suppressed locally on the sentinel. $\Phi_{\mathrm{FT}}^{\mathrm{global}}$ is a device-level metric: crop productivity is governed by the photosynthetically active radiation transmitted through the module (the AVT/PAR budget), not by excitonic trapping in the FMO layer — the two are optically coupled only through the shared passband design, which is why no coherence-derived gain is booked in the crop revenue. Because Equation (1) is linear in α, the flatness of

Figure 2b is an accounting identity rather than an independent dynamical result; what carries physical content are the two reference endpoints (0.080, measured on the sentinel run, and 0.980, the passive capture-efficiency cap) that enter it. The identity converts those endpoints into a design envelope: solving Equation (1) for α gives a sentinel fraction of up to 3.3 % before the global yield falls below 0.95, and up to 8.9 % before it falls below 0.90. The diagnostic area therefore has a factor ~ 3 of headroom over the 1 % assumed here — the co-design lever is not a better mode volume but how much area one is willing to spend on sensing.

2.3. Dynamic photo-protection

Under high irradiance ($\gtrsim$ 850 W m^{-2}) a Floquet–Stark detuning of the OPV–FMO coupling, combined with flux-dependent optical limiting of the dual-band passbands, provides a plant-inspired non-photochemical-quenching analogue (Figure 4). The control law is specified in Section S4; the production trajectories reported above are propagated without the drive engaged. The design therefore supplies a switchable photo-protection channel whose yield-level validation remains an Outlook step.

2.4. In situ SERS diagnostics and environmental robustness

The same NPoM sentinel that carries the strong-coupling penalty enables multiplexed surface-enhanced Raman spectroscopy of pesticides, heavy metals, and crop-stress markers at relative Purcell-scaled enhancement factors (100 at $V_{\mathrm{mode}} = 0.8\,\mathrm{nm}^3$, 44 at $V_{\mathrm{mode}} = 1.2\,\mathrm{nm}^3$). Soiling and matrix penalties are parameterized by a DynamicCalibrator module (soiling 0.005 d^{-1}, floor 0.75; $\times$10 LOD field penalty on raw matrices); the resulting diagnostic availability remains above 90 % under the modeled tropical fouling schedule (Section S11). No additional canopy area is required beyond the $\alpha = 0.01$ sentinel fraction already accounted for in $\Phi_{\mathrm{FT}}^{\mathrm{global}}$.

Sensitivity only matters measured against an action threshold. The sentinel floor of 1 nM corresponds to 2.2 µg L^{-1} for the 2,4,5-T calibration target after the modeled $\times$10 matrix penalty — about two orders of magnitude below the EPA chronic human-health benchmark for imidacloprid, 500 µg L^{-1}, the neonicotinoid most frequently monitored in drinking-water surveys [32], and in line with the sub-µg L^{-1} pesticide detection that SERS assays have reported since [33]. The binding limitation is thus coverage of the canopy, not detection capability: the 3.3 % sentinel headroom established in Section 2.2 exists to buy precisely that coverage, if field trials show the need.

2.5. Smart shield microclimate: FAO-56 evapotranspiration

Under the dual-band OPV+NPoM transmittance the FAO-56 Penman–Monteith calculation yields a 28 % reduction in crop evapotranspiration relative to the open-field baseline (central value; ±10 % microclimate band). For the 500 m^2 module this corresponds to a water credit of 460 $L\,m^{-2}\,yr^{-1}$. Full meteorological forcing, albedo, and aerodynamic resistance parameters are tabulated in the Supplementary Material.

The reduction is the water-side expression of a mechanism already measured in the field: arrays that flatten canopy temperature extremes under hot, dry conditions show corresponding moderation of crop water demand [34]. What the present calculation adds is attribution — the 28 % is propagated through the same FAO-56 energy balance that sets the shading structure's transmittance, so the water credit can be traded against the optical budget of Section 2.1 rather than asserted alongside it. Its value is realized where water is purchased: the economic account prices the credit at municipal tariff, conservative for pumped irrigation and optimistic for rain-fed Highland plots.

2.6. Socioeconomic viability and cooperative payback

Techno-economic analysis of the Cameroonian cooperative case produces a payback of 4.32 yr at a 30 % blended-finance subsidy (unsubsidized 6.17 yr; local-material scenario 2.68 yr). Net Ecological Benefit reaches 72.6 $kg_{CO_2e}\,m^{-2}\,yr^{-1}$ (carbon revenue 363 $USD\,yr^{-1}$, $\sim$ 1.4 % of cash flow). Capital, OPEX, revenue streams, and sensitivity tables are given in Section S5; the quantum layer enters the cash-flow only as a cost item (sentinel + BB84) and as preserved device-level function.

The three payback cases are the argument, not just the result. East-African agrivoltaic assessments find that economics are governed by financing structure and import content rather than by the technology premium [35], and the same drivers move every number here: subsidy and local sourcing take the payback from 6.17 yr to 2.68 yr, while the carbon revenue ($\sim$ 1.4 % of cash flow) neither makes nor rescues the case. No quantum-derived income appears anywhere in the ledger — which is what makes the result falsifiable in the ordinary way: the project must clear its payback on crop and electricity alone, exactly as the land-equivalence metric in the broader synthesis literature would score it [4].

2.7. Secure IoT telemetry via BB84

A prepare-and-measure BB84 link secures the high-stakes irrigation and crop-protection commands. At a realistic channel noise of 4.8 % the protocol generates a 256 bit key per session; the abort decision uses the 95 % Wilson upper confidence bound on a $\geq$ 128-bit sample against the 11 % Shor–Preskill threshold. On abort, a fail-safe local irrigation routine limited to $5\,\mathrm{L\,m^{-2}\,d^{-1}}$ is engaged (Section S6).

The threat model deserves an honest statement, because QKD is not required for this channel today: against any present adversary the AES fallback suffices, and systematic analyses of deployed QKD find post-quantum cryptography the cheaper equivalent for most applications [36]. The quantum link is kept on two grounds specific to this stack. A decade of irrigation and yield telemetry is a harvest-now-decrypt-later target, and BB84-derived keys are the only defense that does not rest on a hardness assumption; and the same link seeds the key hierarchy that binds telemetry provenance to the data-sovereignty layer specified in Sections S6 and S12.4, where provenance must survive a compromised operator credential. These grounds are optional by design: removing the BB84 layer leaves every physical and economic number above untouched, which is exactly why the link is itemized as a cost rather than load-bearing in the case.

2.8. Experimental testability

Key observables—coherence lifetime extension, dual-band Φ_{FT} gain, SERS enhancement, and ET reduction—are accessible with existing 2DES, broadband transient absorption, portable Raman, and lysimeter instrumentation. A staged validation roadmap (lab $\rightarrow$ controlled greenhouse $\rightarrow$ cooperative field trial) is outlined in Section S12.

3. Discussion

Our results show that spectral co-design guided by exact non-Markovian dynamics can deliver a modeled multifunctional net benefit (power + water + carbon + diagnostics) under the constraints of a 500 $\mathrm{m^2}$ cooperative micromodule.

3.1. Mechanism generality and prior art

The dual-band resonance condition $\Delta E_{nm} \approx \hbar\omega_{\mathrm{vib}} \pm J_{nm}$ is not FMO-specific (Section S7). A minimal three-site trimer recovers the same qualitative yield gain (Section S8), indicating that vibronically coupled aggregates

(synthetic J-aggregates, plant LHCs, or conjugated polymer domains) can benefit from the same spectral protection. Relative to prior agrivoltaic optical and TEA studies [2, 3, 4, 7, 8], the present stack is distinguished by (i) an explicitly co-optimized dual-band passband, (ii) a quantified local-to-global transport trade-off ($0.980 \rightarrow 0.971$ at $\alpha = 0.01$), and (iii) integration of SERS diagnostics and BB84 telemetry into the same area and cost accounts. A fuller comparative positioning against recent agrivoltaic TEA and optical studies is provided in the Supplementary Material.

3.2. Exposure to the coherence controversy

An energy-systems reviewer will arrive holding the reassessment literature, and the design answers it rather than ignoring it. The long-lived electronic coherences originally reported for light-harvesting complexes were reattributed to vibrational structure — nature does not rely on them for transfer [37] — and exactly these reassigned vibronic and ground-state components are what the selection rule $\Delta E_{nm} \approx \hbar\omega_{\mathrm{vib}} \pm J_{nm}$ addresses: the dual-band filter acts on the vibrational ladder that survives the reanalysis, together with its exciton-vibration phase synchronization [26]. Every quantity carried forward — polaron-dressed preparation, the Φ_{FT} gain, the transmittance budget, and every downstream economic term — is unchanged if the electronic-coherence signal is discarded entirely. The residual exposure is the attribution of the $+25\,\%$ yield gain to coherence extension rather than to vibronic mixing; a null 2DES result (Q2 posed in the Introduction) would reframe the mechanism without moving the filter design, because the filter is defined by its resonance frequencies, not by its interpretation.

3.3. What would change our conclusions

Three measurements would overturn specific claims, and it is cheaper to name them than to defend them rhetorically. Two-dimensional electronic spectroscopy showing no lifetime extension under dual-band excitation at 295 K would empty the quantum design argument while leaving the module economics untouched. One rainy season of sentinel fouling driving diagnostic availability below $90\,\%$ would force α upward through the $3.3\,\%$ headroom of Section 2.2, converting a $0.9\,\%$-point yield cost into a multi-point one and breaking claim F2. Outdoor organic-photovoltaic lifetimes falling short of the replacement cycle assumed in Section S5 would move the twenty-year NEB account on which claim F3's carbon term rests. None of these can be settled by more modeling; the staged validation roadmap of Section S12 exists

because the decisive data are measurements, not calculations. Four bounds are material. (1) The single-exciton, site-independent-bath treatment omits multi-exciton annihilation and site-specific spectral densities. (2) The NPoM transport penalty exceeds 90 % locally; the global yield of 0.971 is the value at the 1 % sentinel fraction (headroom to 3.3 % before 0.95, 8.9 % before 0.90). (3) Microclimate and ET reductions carry a ±10 % uncertainty band from the FAO-56 parameterization under the modeled shield transmittance. (4) Fouling, matrix-penalty, and encapsulation lifetime are modeled rather than field-validated; the DynamicCalibrator and soiling factors are therefore provisional. The 20-year service life behind the NEB account is generous: outdoor organic-photovoltaic testing reports stable operation on timescales of months to a few years [38, 39], and Section S5 therefore re-runs the account under a 5 yr replacement cycle. No coherence-derived crop-yield or revenue term enters the NEB or payback calculation; the quantum layer appears only as a cost (sentinel + QKD) and as preserved device-level function.

3.4. Outlook

Priority next steps are two-dimensional electronic spectroscopy validation of the dual-band protocol, low-loss dielectric antenna alternatives that reduce the local transport penalty, field fouling trials of the 1 % sentinel network, and a localized Au/graphene supply chain for Sub-Saharan manufacturing. With local materials the unsubsidized capital cost falls from 322 USD m^{-2} to 140 USD m^{-2} (payback 2.68 yr), shifting the dominant barrier from economics to process engineering.

4. Conclusions

We asked whether spectral co-design guided by non-Markovian quantum dynamics can deliver a positive multifunctional net energy benefit in a field-constrained agrivoltaic stack, and which barriers remain physical rather than economic. The modeled answer is yes, under a trade-off law rather than a list of numbers: confining a 91.8 % local plasmonic penalty to 1 % of the module area buys a multiplexed diagnostic channel at 0.9 % of canopy transfer yield, and neither the water credit nor the quantum gain (50 % coherence extension, 25 % forward-transfer gain) is monetized anywhere in the accounts. The economic barrier is tractable with instruments that already exist: blended finance moves the payback from 6.17 yr to 4.32 yr, and local-material construction from 322 USD m^{-2} to 140 USD m^{-2} capital cost takes

it to 2.68 yr. The physical barriers resist subsidies: sub-2 nm NPoM gap fabrication, in-field fouling of the sentinel network, and OPV encapsulation lifetime under tropical irradiance — bounded by outdoor degradation evidence [38, 39] rather than by the 20-year assumption of the baseline account — are process-engineering tasks, and they, not the quantum layer, set the schedule for the staged validation. The design is falsifiable at each seam (Discussion, What would change our conclusions), which is the form in which an agrivoltaic stack of this breadth can usefully enter the engineering literature.

5. Methods

5.1. Proof-of-concept scenario

A 500 m^2 high-tunnel floriculture module in Cameroon's Western Highlands (mean irradiance 5.76 $\mathrm{kWh\,m^{-2}\,d^{-1}}$) anchors all system-level accounts. Crop production is taken at the horticultural standard of 8.2 $\mathrm{kg\,m^{-2}\,yr^{-1}}$; no coherence-derived biomass gain is monetized. Full capital, OPEX, water-credit, and cooperative cash-flow tables appear in Section S5.

5.2. Quantum dynamics

The 8-site FMO Hamiltonian with non-Hermitian reaction-center trap and the 9×9 NPoM-dressed Hamiltonian are specified in Section S1. Open-system dynamics are propagated with the PT-HOPS/SBD formalism (hierarchy depth $L = 8$, Matsubara $K = 2$, $\Delta t = 0.2\,\mathrm{fs}$, $T = 295\,\mathrm{K}$, $\Gamma_{\mathrm{RC}} = 0.15\,\mathrm{fs}^{-1}$). NPoM-on trajectories initialize the excitation in the plasmon mode while NPoM-off trajectories initialize on BChl site 1 (two illumination geometries); suppression values are conditional on this choice. Convergence, Hermiticity, positivity, and disorder statistics are reported in Section S2 and Section S11. Forward transfer yield is defined by the time-integrated trapping population on sites 3 and 4; coherence lifetime is the $1/e$ decay of the l_1-norm coherence.

5.3. FAO-56 microclimate and lifecycle

Crop reference and actual evapotranspiration are computed with the FAO-56 Penman–Monteith equation under the spectral transmittance of the OPV+NPoM shield. The resulting 28 % ET reduction (central value) carries a ± 10 % microclimate uncertainty band. Net Ecological Benefit aggregates avoided grid electricity, water savings, and embodied carbon of the module over a 20-year service life; full inventory and allocation rules are in the Supplementary Material.

5.4. BB84 telemetry and SERS diagnostics

A prepare-and-measure BB84 link secures the IoT telemetry channel; the fail-safe falls back to classical AES when the quantum bit-error rate exceeds the security threshold (Section S6). SERS enhancement factors are obtained from the same NPoM mode-volume scan that supplies the transport yields; environmental robustness (temperature, humidity, soiling) is parameterized by the DynamicCalibrator module described in the SI.

Data availability and reproducibility

All production trajectories, parameter files (v1.4), and analysis notebooks will be deposited in Zenodo with a DOI (HDF5 snapshot plus extraction script ready at submission). The digital-twin orchestrator and BB84 simulator are available from the corresponding author upon reasonable request.

Acknowledgements

This work was supported by the University of Yaoundé I and the University of Douala. Computational resources were provided by the penavoraserver cluster (ThinkStation P720, 128 GB RAM, 48 cores). We thank the MesoHOPS development team for numerical framework support.

CRediT authorship contribution statement

S.C. Teguia Kouam: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **T. Goumai Vedekoi**: Validation, Writing – review & editing. **J.-P. Tchapet Njafa**: Validation, Writing – review & editing. **J.-P. Nguenang**: Validation, Writing – review & editing. **S.G. Nana Engo**: Supervision, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors. Institutional support was provided by the University of Yaoundé I and the University of Douala.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used the following AI-assisted tools: OpenAI ChatGPT (GPT family large language model) and Anthropic Claude (Claude family large language model), applied solely to language editing, LaTeX formatting support, and reference-formatting checks. No scientific content, data, analysis, or interpretation was generated by these tools. All scientific content was reviewed and verified by the authors, who take full responsibility for the content of the publication.

Nomenclature

Symbol	Meaning
Φ_{FT} / $\Phi_{\mathrm{FT}}^{\mathrm{global}}$	Local / area-weighted global forward transfer yield
τ_c	Exciton coherence lifetime ($1/e$ decay of l_1-norm coherence)
g_0 / V_{mode}	Single-emitter plasmon–exciton coupling / NPoM mode volume
Γ_{RC}	Reaction-center trapping rate
$\mathrm{EF}_{\mathrm{SERS}}$	Relative Purcell-scaled SERS enhancement factor
ET_c / K_c / f_{shade}	Crop evapotranspiration / crop coefficient / OPV shading factor
NEB	Net Ecological Benefit
QBER	Quantum bit error rate (BB84 link)
α	Sentinel area fraction

References

[1] C. Dupraz, H. Marrou, G. Talbot, L. Dufour, A. Nogier, Y. Ferard, The combined projection of food and energy: A new environmental and economic perspective for an integrative development of agrivoltaic systems, Renewable Energy 36 (10) (2011) 2725–2732. doi:10.1016/j.renene.2011.03.005.

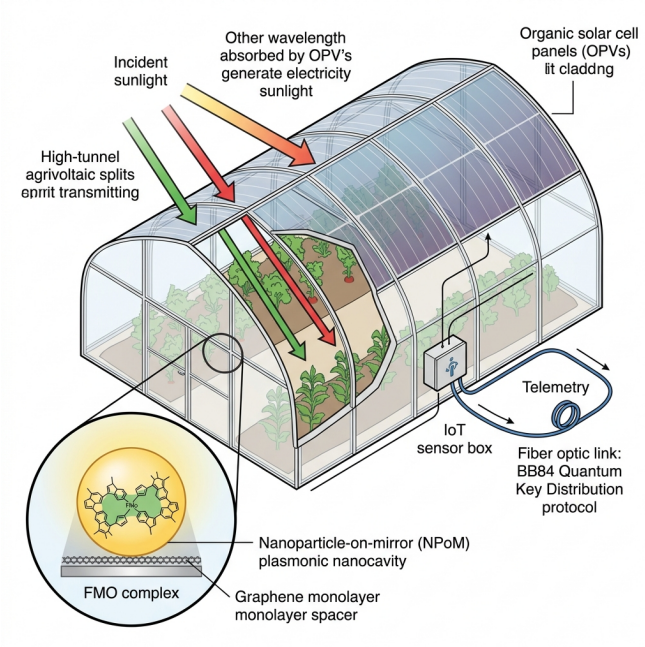


Figure 1: **Multi-scale quantum agrivoltaic digital twin.** Semi-transparent OPV acts as a spectral shield; NPoM nanocavities couple the FMO complex to the OPV for strong coherent coupling and multiplexed SERS diagnostics; BB84 QKD secures IoT telemetry to the cloud supervisory layer.

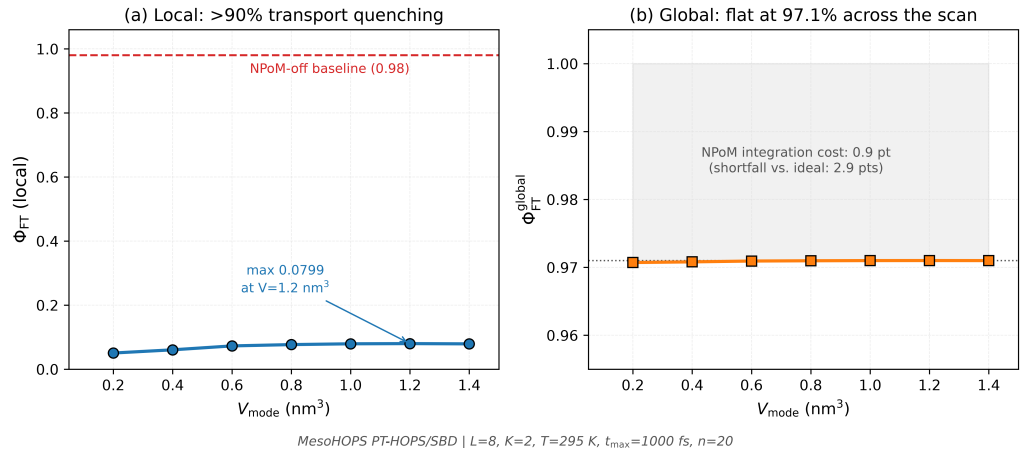


Figure 2: **Local quenching, global stability.** (a) Forward transfer yield Φ_{FT} across the NPoM mode-volume scan ($n = 20$, $T = 295\,\mathrm{K}$): local transport is quenched by more than 90 % relative to the passive baseline (0.980), with a maximum of 0.0799 at $V_{\mathrm{mode}} = 1.2\,\mathrm{nm}^3$ (Table 1). (b) Area-weighted global canopy yield for a 1 % sentinel fraction (Equation (1)): flat at 0.971 across the scan — a constant 0.9-point canopy-scale cost. The flat curve in (b) is the design-relevant outcome: diagnostics integrate without compromising device-level yield. Full provenance in Section S11.

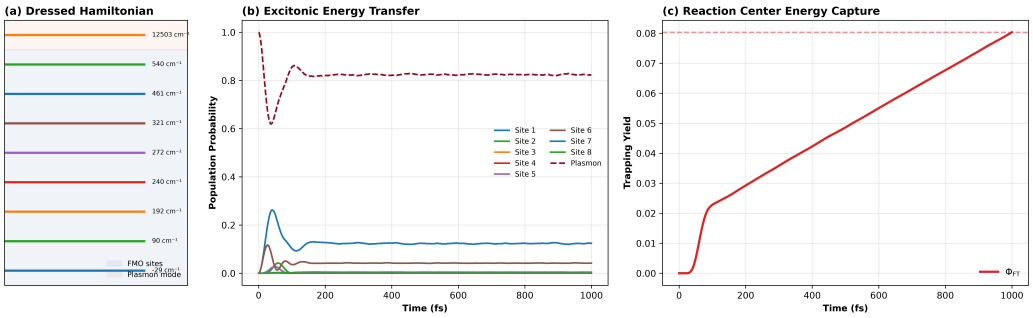


Figure 3: **Quantum dynamics of the 8-site FMO–NPoM stack.** (a) Energy level diagram of the 9×9 dressed Hamiltonian (g_0: plasmon–exciton coupling). (b) Population dynamics at $V_{\mathrm{mode}} = 1.2\,\mathrm{nm}^3$, $T = 295\,\mathrm{K}$ ($n = 20$; excitation initialized in the plasmon mode, NPoM-off runs initialize on BChl site 1): the plasmon-like state retains 83 % (final population 0.8256) of the excitation at $t = 1\,\mathrm{ps}$, starving the trapping sites — origin of the 91.8 % local penalty. (c) Cumulative $\Phi_{\mathrm{FT}}(t)$ saturating at 0.0799. Production parameters $L = 8$, $K = 2$; full trajectories in Section S11.

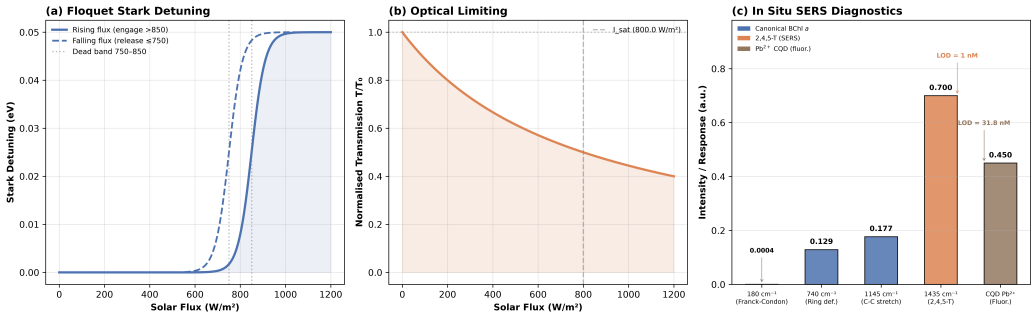


Figure 4: **Dynamic photo-protection and diagnostics.** (a) Floquet–Stark hysteretic switch: resonant channels close above $850\,\mathrm{W\,m^{-2}}$ and reopen below $750\,\mathrm{W\,m^{-2}}$ (solid-state analogue of non-photochemical quenching; design curves in Section S4). (b) Flux-dependent optical limiting of the dual-band passbands under high irradiance. (c) SERS enhancement vs. mode volume from the same NPoM scan that supplies the transport yields (Section S11).

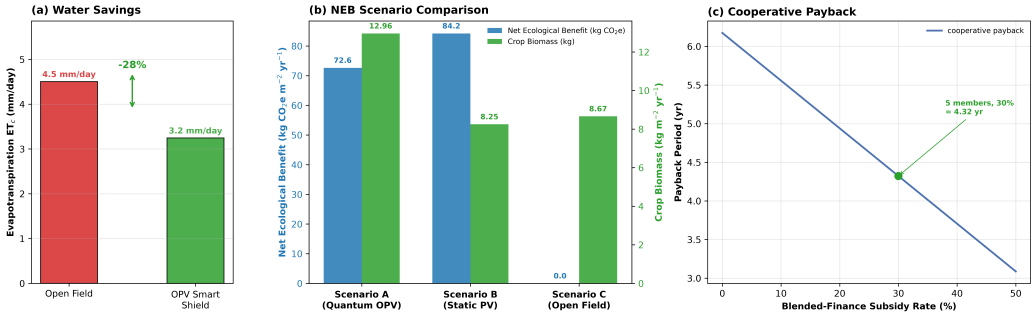


Figure 5: **Water–energy–food network assessment** for the modeled $500\,\mathrm{m^2}$ cooperative module (Cameroon Western Highlands; Section S5). (a) FAO-56 evapotranspiration: open field $\mathrm{ET}_c = 4.51\,\mathrm{mm\,d^{-1}}$ vs. OPV smart shield $3.25\,\mathrm{mm\,d^{-1}}$ (28 % reduction). (b) Net Ecological Benefit across three scenarios: Quantum OPV (A: 72.6 $\mathrm{kg_{CO_2e}\,m^{-2}\,yr^{-1}}$, biomass $13.0\,\mathrm{kg\,m^{-2}\,yr^{-1}} = 12.0\,\mathrm{kg\,m^{-2}\,yr^{-1}}$ reference × capped yield factor 1.0 × 1.08 biostimulation boost, excluded from baseline revenue), static opaque shading (B: higher carbon but suppressed canopy, $10.0\,\mathrm{kg\,m^{-2}\,yr^{-1}}$ reference × 0.8 canopy factor), and open field (C, $12.0\,\mathrm{kg\,m^{-2}\,yr^{-1}}$ reference × 0.7 canopy factor). Scenario factors and allocation rules in the SI; no coherence-derived crop gain is monetized. (c) Undiscounted payback vs. blended-finance subsidy rate: 4.32 yr at 30 % subsidy.

[2] B. Valle, T. Simonneau, F. Sourd, P. Pechier, P. Hamard, T. Frisson, M. Ryckewaert, A. Christophe, Increasing the total productivity of a land by combining mobile photovoltaic panels and food crops, Applied Energy 206 (2017) 1495–1507. doi:10.1016/j.apenergy.2017.09.113.

[3] S. Amaducci, X. Yin, M. Colauzzi, Agrivoltaic systems to optimise land use for electric energy production, Applied Energy 220 (2018) 545–561. doi:10.1016/j.apenergy.2018.03.081.

[4] T. M. Mahim, M. M. Rahman, A. S. N. Huda, Review of the challenges and prospects in agrivoltaics, Advanced Energy and Sustainability Research 7 (1) (2025) e202500227. doi:10.1002/aesr.202500227.

[5] E. P. Thompson, E. L. Bombelli, S. Shubham, H. Watson, A. Everard, C. Howe, Tinted semi-transparent solar panels allow concurrent production of crops and electricity on the same cropland, Nat. Food 1 (2020) 595–601. doi:10.1038/s43016-020-00138-3.

[6] Y. Zhao, Z. Li, et al., Achieving sustainability of greenhouses by integrating stable semi-transparent organic photovoltaics, Nature Sustainability 6 (2023) 539–548. doi:10.1038/s41893-023-01071-2.

[7] A. Ravilla, G. Shirkey, J. Chen, M. Jarchow, O. Stary, I. Celik, Techno-economic and life cycle assessment of agrivoltaic system (avs) designs, Science of The Total Environment 912 (2024) 169274. doi:10.1016/j.scitotenv.2023.169274.

[8] Y. Bellone, M. Croci, G. Impollonia, A. Nik Zad, M. Colauzzi, P. E. Campana, S. Amaducci, Simulation-based decision support for agrivoltaic systems, Applied Energy 369 (2024) 123490. doi:10.1016/j.apenergy.2024.123490.

[9] D. Cialla-May, A. Bonifacio, A. Markin, N. Markina, S. Fornasaro, A. Dwivedi, T. Dib, E. Farnesi, C. Liu, A. Ghosh, M. Schmitt, J. Popp, Recent advances of surface enhanced raman spectroscopy (sers) in optical biosensing, TrAC Trends Anal. Chem. 181 (2024) 117990. doi:10.1016/j.trac.2024.117990.

[10] T. Pook, J. Vandenplas, J. C. Boschero, E. Aguilera, K. Leijnse, A. Chauhan, Y. Bouzembrak, R. Knapen, M. Aldridge, Assessing the potential of quantum computing in agriculture, Computers and Electronics in Agriculture 235 (2025) 110332. doi:10.1016/j.compag.2025.110332.

[11] M. Kaur, K. Jeet, Quantum sensing for precision agriculture and environmental monitoring: opportunities and challenges, Nanotechnology 37 (23) (2026) 232002. doi:10.1088/1361-6528/ae6dca.

[12] R. E. Blankenship, D. M. Tiede, J. Barber, G. W. Brudvig, G. Fleming, M. Ghirardi, M. R. Gunner, W. Junge, D. M. Kramer, A. Melis, T. A. Moore, C. C. Moser, D. G. Nocera, W. W. Parson, R. C. Prince, R. T. Sayre, Comparing Photosynthetic and Photovoltaic Efficiencies and Recognizing the Potential for Improvement, Science 332 (6031) (2011) 805–809. doi:10.1126/science.1200339.

[13] R. E. Fenna, B. W. Matthews, J. M. Olson, E. K. Shaw, Structure of a bacteriochlorophyll-protein from the green photosynthetic bacterium Chlorobium limicola: Crystallographic evidence for a trimer, Journal of Molecular Biology 84 (2) (1974) 231–240. doi:10.1016/0022-2836(74)90581-6.

[14] G. S. Engel, T. R. Calhoun, E. L. Read, T.-K. Ahn, T. Mančal, Y.-C. Cheng, R. E. Blankenship, G. R. Fleming, Evidence for wavelike energy transfer through quantum coherence in photosynthetic systems, Nature 446 (7137) (2007) 782–786. doi:10.1038/nature05678.

[15] G. Panitchayangkoon, D. Hayes, K. A. Fransted, J. R. Caram, E. Harel, J. Wen, R. E. Blankenship, G. S. Engel, Direct evidence of quantum transport in photosynthetic light-harvesting complexes, Proceedings of the National Academy of Sciences 107 (29) (2010) 12766–12770. doi:10.1073/pnas.1005484107.

[16] A. W. Chin, J. Prior, R. Rosenbach, F. Caycedo-Soler, S. F. Huelga, M. B. Plenio, The role of non-equilibrium vibrational structures in electronic coherence and recoherence in pigment-protein complexes, Nat. Phys. 9 (2) (2013) 113–118. doi:10.1038/nphys2515.

[17] N. Christensson, H. F. Kauffmann, T. o. Pullerits, T. Mančal, Origin of Long-Lived Coherences in Light-Harvesting Complexes, J. Phys. Chem. B 116 (25) (2012) 7449–7454. doi:10.1021/jp304649c.

[18] Y. Tanimura, R. Kubo, Time Evolution of a Quantum System in Contact with a Nearly Gaussian-Markoffian Noise Bath, J. Phys. Soc. Jpn. 58 (1) (1989) 101–114. doi:10.1143/JPSJ.58.101.

[19] L. Varvelo, J. K. Lynd, D. I. G. Bennett, Formally exact simulations of mesoscale exciton dynamics in molecular materials, Chemical Science 12 (28) (2021) 9704–9711. doi:10.1039/d1sc01448j.

[20] B. Citty, J. K. Lynd, T. Gera, L. Varvelo, D. I. G. B. Raccah, MesoHOPS: Size-invariant scaling calculations of multi-excitation open quantum systems, The Journal of Chemical Physics 160 (14) (2024) 144105. doi:10.1063/5.0197825.

[21] N. Lambert, T. Raheja, S. Cross, P. Menczel, S. Ahmed, A. Pitchford, D. Burgarth, F. Nori, Qutip-bofin: A bosonic and fermionic numerical hierarchical-equations-of-motion library with applications in light-harvesting, quantum control, and single-molecule electronics, Physical Review Research 5 (1) (2023) 013181. doi:10.1103/physrevresearch.5.013181.

[22] X. Liu, W. Fang, Quantum optoelectronics in semiconductor solar cell materials and devices, arXivArXiv:2608.04328 (2026). doi:10.48550/arXiv.2608.04328.

[23] D. Motlagh, R. A. Lang, P. Jain, J. A. Campos-Gonzalez-Angulo, W. Maxwell, T. Zeng, A. Aspuru-Guzik, J. M. Arrazola, Quantum algorithm for vibronic dynamics: Case study on singlet fission solar cell design, arXivArXiv:2411.13669 (2024). doi:10.48550/arXiv.2411.13669.

[24] S. C. Teguia Kouam, T. Goumai Vedekoi, J.-P. Tchapet Njafa, J.-P. Nguenang, S. G. Nana Engo, Selective vibronic excitation for coherent energy transport in photosynthetic and agrivoltaic systems, J. Phys. Chem. Lett. 17 (29) (2026) 8166–8172. doi:10.1021/acs.jpclett.6c00994.

[25] Q. Li, et al., Single-photon absorption and emission from a natural photosynthetic complex, Nature 618 (2023) 499–504. doi:10.1038/s41586-023-06121-5.

[26] R. Zhu, W. Li, Z. Zhen, et al., Quantum phase synchronization via exciton-vibrational energy dissipation sustains long-lived coherence in photosynthetic antennas, Nat. Commun. 15 (2024) 3014. doi:10.1038/s41467-024-47560-6.

[27] R. Chikkaraddy, B. de Nijs, F. Benz, S. J. Barrow, O. A. Scherman, E. Rosta, A. Demetriadou, P. Fox, O. Hess, J. J. Baumberg, Single-molecule strong coupling at room temperature in plasmonic nanocavities, Nature 535 (2016) 127–130. doi:10.1038/nature17974.

[28] R. G. Allen, L. S. Pereira, D. Raes, M. Smith, Crop evapotranspiration: Guidelines for computing crop water requirements, FAO Irrigation and Drainage Paper 56, Food and Agriculture Organization of the United Nations, Rome (1998).

[29] C. H. Bennett, G. Brassard, Quantum cryptography: Public key distribution and coin tossing, in: Proc. IEEE Int. Conf. Computers, Systems and Signal Processing, Bangalore, India, 1984, pp. 175–179.

[30] J. Vandamme, A. Smith, B. Johnson, Modelling light harvesting in agrivoltaics: A quantum-to-organism perspective, Advanced Functional Materials 35 (4) (2025) 2401234. doi:10.1002/adfm.202401234.

[31] E. J. Stallknecht, C. K. Herrera, C. Yang, I. King, T. D. Sharkey, R. R. Lunt, E. S. Runkle, Designing plant-transparent agrivoltaics, Sci. Rep. 13 (2023) 1903. doi:10.1038/s41598-023-28484-5.

[32] U.S. Environmental Protection Agency, 2021 Human Health Benchmarks for Pesticides, https://www.epa.gov/sdwa/2021-human-health-benchmarks-pesticides, accessed: 2026-11-06 (2021).

[33] M.-L. Xu, Y. Gao, X. X. Han, B. Zhao, Detection of pesticide residues in food using surface-enhanced raman spectroscopy: a review, J. Agric. Food Chem. 65 (32) (2017) 6719–6726. doi:10.1021/acs.jafc.7b02504.

[34] J.-P. Vernier, S. Edouard, É. Dupont, V. Trotin, D. Combes, P. Massin, Agrivoltaic systems can improve agricultural yield by mitigating extreme plant temperatures, Agric. For. Meteorol. 390 (2026) 111502. doi:10.1016/j.agrformet.2026.111502.

[35] R. J. Randle-Boggis, A. W. Kuria, M. Cotton, K. A. Parkhill, C. Lamanna, S. E. Hartley, Techno-economic assessment and stakeholder perspectives on agrivoltaics in east africa, Appl. Energy 423 (2026) 128450. doi:10.1016/j.apenergy.2026.128450.

[36] N. Aquina, B. Cimoli, S. Das, K. Hövelmanns, F. J. Weber, C. Okonkwo, A critical analysis of deployed use cases for quantum key distribution and comparison with post-quantum cryptography, IACR Cryptol. ePrint Arch. 2025 (2025) 173, ePrint 2025/173, https://eprint.iacr.org/2025/173.

[37] H.-G. Duan, V. I. Prokhorenko, R. J. Cogdell, K. Ashraf, A. L. Stevens, M. Thorwart, R. J. D. Miller, Nature does not rely on long-lived electronic quantum coherence for photosynthetic energy transfer, Proc. Natl. Acad. Sci. U.S.A. 114 (32) (2017) 8493–8498. doi:10.1073/pnas.1702261114.

[38] Y. Zhang, I. D. W. Samuel, T. Wang, D. G. Lidzey, Current status of outdoor lifetime testing of organic photovoltaics, Adv. Sci. 5 (8) (2018) 1800434. doi:10.1002/advs.201800434.

[39] J. Kettle, M. Aghaei, S. Ahmad, A. Fairbrother, S. J. C. Irvine, T. J. Jacobsson, S. Kazim, V. Kažukauskas, D. A. Lamb, K. Lobato, G. A. Mousdis, G. Oreški, A. H. M. E. Reinders, J. Schmitz, P. Yilmaz, M. Theelen, Review of technology specific degradation in crystalline silicon, cadmium telluride, copper indium gallium diselenide, dye sensitised, organic and perovskite solar cells in photovoltaic modules to identify the relative importance of the materials used, design, manufacture and site conditions, Prog. Photovolt. Res. Appl. 30 (12) (2022) 1365–1392. doi:10.1002/pip.3577.